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Categorical and Numeric Variables

Q1) a) 4 unique labels

column order

[red, blue, green, grey]

∴ one hot encoded matrix is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b)

$$\hat{p}_k = \frac{\text{number of observations in category } k}{n}$$

$$\therefore \hat{p}_{\text{red}} = \frac{2}{6} = \frac{1}{3}, \hat{p}_{\text{blue}} = \frac{1}{3}, \hat{p}_{\text{green}} = \frac{1}{6},$$

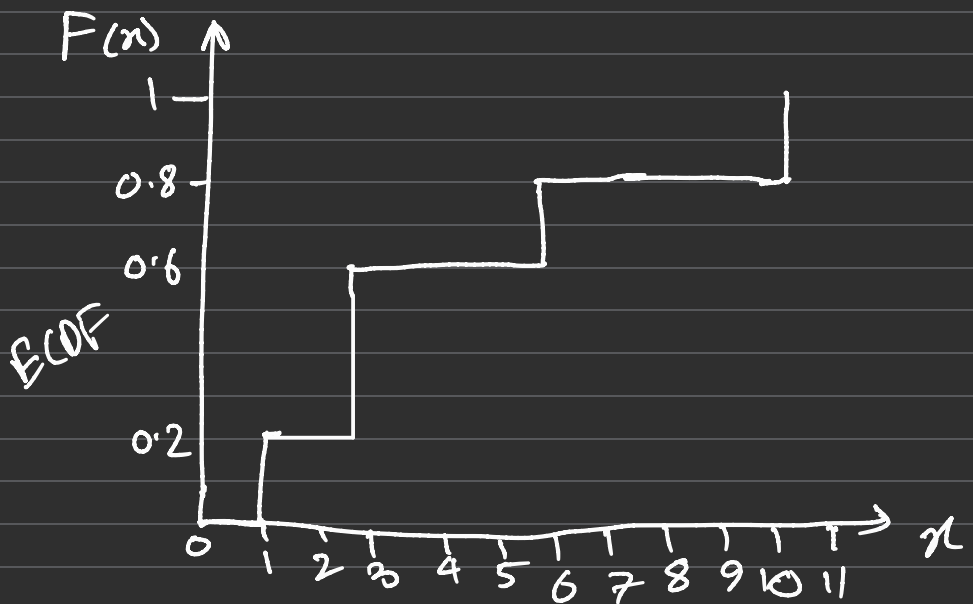
$$\hat{p}_{\text{grey}} = \frac{1}{6}$$

c)

$$X = [1, 3, 3, 6, 11], \quad n = 5$$

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(x_i \leq x)$$

$$\therefore \hat{F}(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{5}, & 1 \leq x < 3 \\ \frac{3}{5}, & 3 \leq x < 6 \\ \frac{4}{5}, & 6 \leq x < 11 \\ 1, & x \geq 11 \end{cases}$$



2a)

$$\begin{aligned} \text{CDF: } F_Y(y) &= P(Y \leq y) \\ &= P(a + bX \leq y) \end{aligned}$$

$\therefore b > 0$, direction of the inequality

stays the same

$$P\left(X \leq \frac{y-a}{b}\right)$$

$$\therefore F_Y(y) = F_X\left(\frac{y-a}{b}\right)$$

$$f_Y(y) = \frac{d}{dy} F_X\left(\frac{y-a}{b}\right)$$

$$= \frac{1}{b} f_X\left(\frac{y-a}{b}\right)$$

$$b) \quad \bar{X} = \frac{1+3+4}{3} = \frac{8}{3} \approx 2.667$$

$$c) \quad X^2 = [1, 9, 16]$$

$$\bar{X^2} = \frac{1+9+16}{3} = \frac{26}{3} \approx 8.667$$

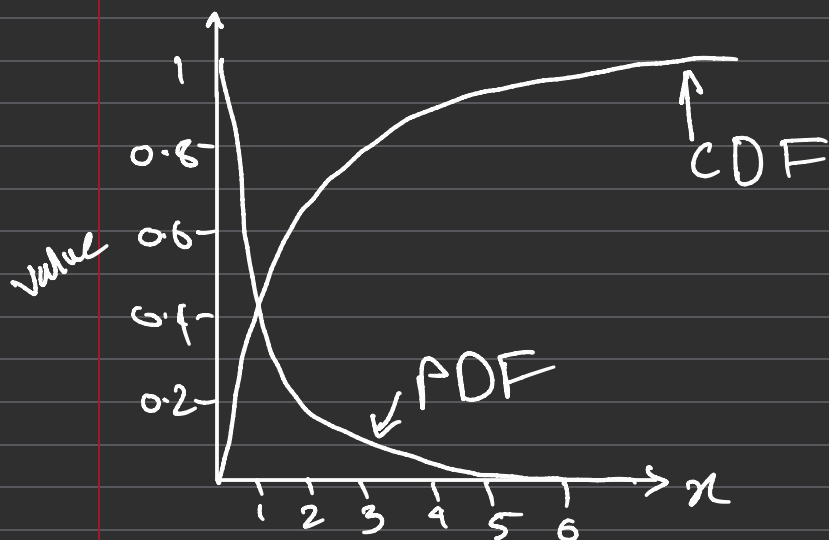
$$d) \quad (\bar{X})^2 = \left(\frac{8}{3}\right)^2 = \frac{64}{9} \approx 7.111$$

$\bar{X^2}$ does not equal to the mean of square from part c.

3a)

$$f_x(x) = \frac{d}{dx} (1 - e^{-x}) = e^{-x}$$

$$\therefore f_x(x) = \begin{cases} 0 & x < 0 \\ e^{-x} & x \geq 0 \end{cases}$$



$$b) P(X \geq 3) = 1 - P(X \leq 3)$$

$$\therefore F_x(3) = 1 - e^{-3}$$

$$P(X \geq 3) = 1 - (1 - e^{-3}) = e^{-3}$$

$$\approx 4.98\%$$

$$P(X \leq 35) = F_X(35)$$

$$\therefore P(X \leq 35) = 1 - e^{-35}$$

$$\approx 1 \approx 100\%$$

c)

From Q2

$$F_Y(y) = F_X\left(\frac{y-a}{b}\right)$$

$$\therefore F_Y(y) = F_X\left(\frac{y-1}{3}\right)$$

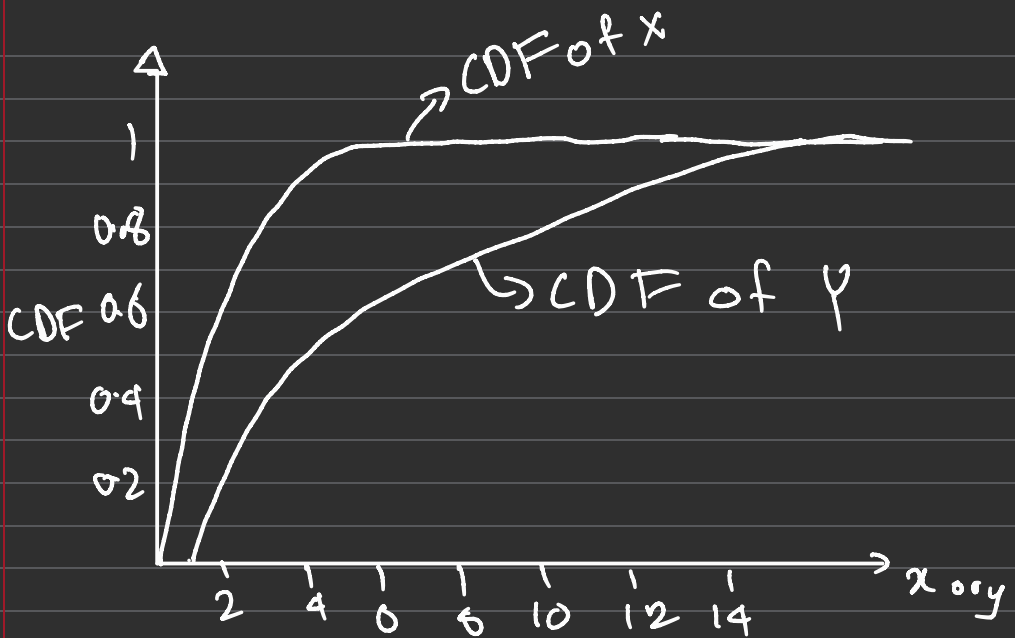
Now substituting CDF

$$F_X(x) = 1 - e^{-x}$$

$$\text{so } F_Y(y) = \begin{cases} 0 & y < 1 \\ 1 - e^{-(y-1)/3}, & y \geq 1 \end{cases}$$

started at $y=1$ because $X \geq 0$

$$\therefore Y = 1 + 3X \geq 1$$



$$P(Y < 10) = P(1 + 3X < 10)$$

$$1 + 3X < 10$$

$$X < 3$$

$$\therefore P(Y < 10) = P(X < 3) = F_X(3)$$

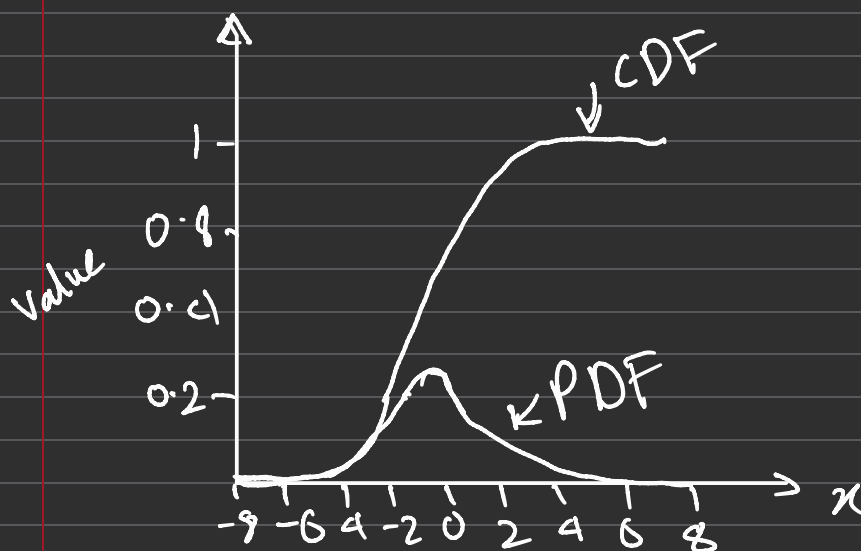
$$= 1 - e^{-3} \approx 0.9502$$

4a)

PDF

$$f_X(x) = \frac{d}{dx} \left(\frac{1}{1+e^{-x}} \right)$$

$$= \frac{e^{-x}}{(1+e^{-x})^2}$$



b)

$$P(X \geq 0.8) = 1 - F_X(0.8)$$

$$= 1 - \frac{1}{1+e^{-0.8}} \approx 0.310$$

$$\approx 31\%$$

$$P(X \leq 0.3) = F_X(0.3) = \frac{1}{1 + e^{-0.3}}$$

$$\approx 0.5744$$

$$\approx 57.44\%$$

$$F_Y(y) = P(Y \leq y)$$

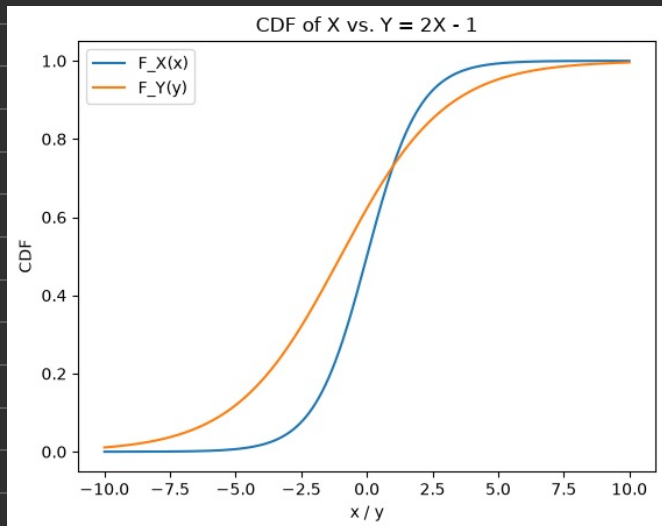
$$Y = 2X - 1$$

$$F_Y(y) = P(2X - 1 \leq y)$$

$$X \leq \frac{y+1}{2}$$

$$\therefore F_Y(y) = F_X\left(\frac{y+1}{2}\right)$$

$$= \frac{1}{1 + e^{-(y+1)/2}}$$



$$P(Y < 0)$$

$$P(2X - 1 < 0)$$

$$P(X < 0.5)$$

$$= F_X(0.5)$$

$$P(Y < 0) = \frac{1}{1 + e^{-0.5}}$$

$$\approx 0.6225$$

$$\approx 62.25\%$$

$$5a) F(x) = 1 - e^{-x}, x \geq 0$$

For median m

$$F(x) = \frac{1}{2}$$

$$\therefore 1 - e^{-m} = \frac{1}{2}$$

$$\Rightarrow -m = \ln\left(\frac{1}{2}\right)$$

$$\therefore m = \ln 2 \approx 0.693$$

$$b) \quad F(x) = \frac{1}{1+e^{-x}}$$

$$\Rightarrow \frac{1}{1+e^{-m}} = \frac{1}{2}$$

$$e^{-m} = 1$$

$$-m = \ln(1) = 0$$

$$\therefore m = 0$$

$$c) \quad F(x) = u$$

$$1 - e^{-x} = u$$

$$e^{-x} = 1 - u$$

$$-x = \ln(1 - u)$$

$$\therefore F^{-1}(u) = -\ln(1 - u), \quad 0 < u < 1$$

$$d) \quad F(u) = u$$

$$\frac{1}{1+e^{-x}} = u$$

$$1 = u + u e^{-x}$$

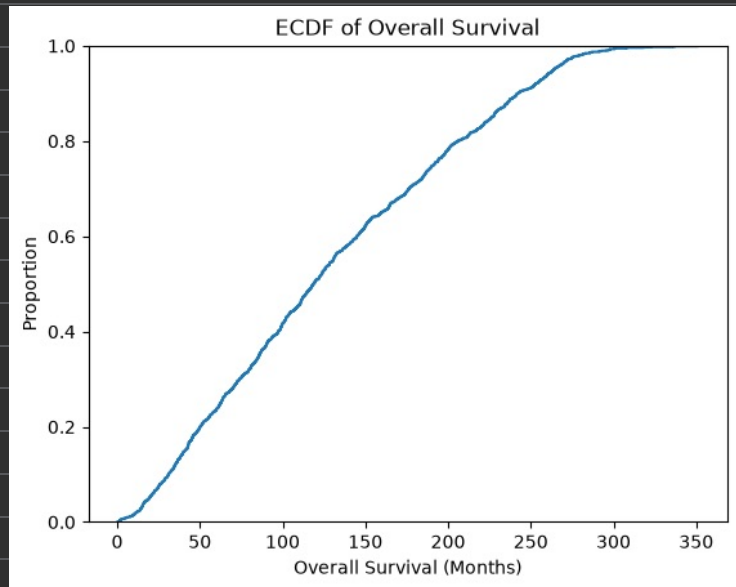
$$e^{-x} = \frac{1-u}{u}$$

$$-x = \ln \left(\frac{1-u}{u} \right)$$

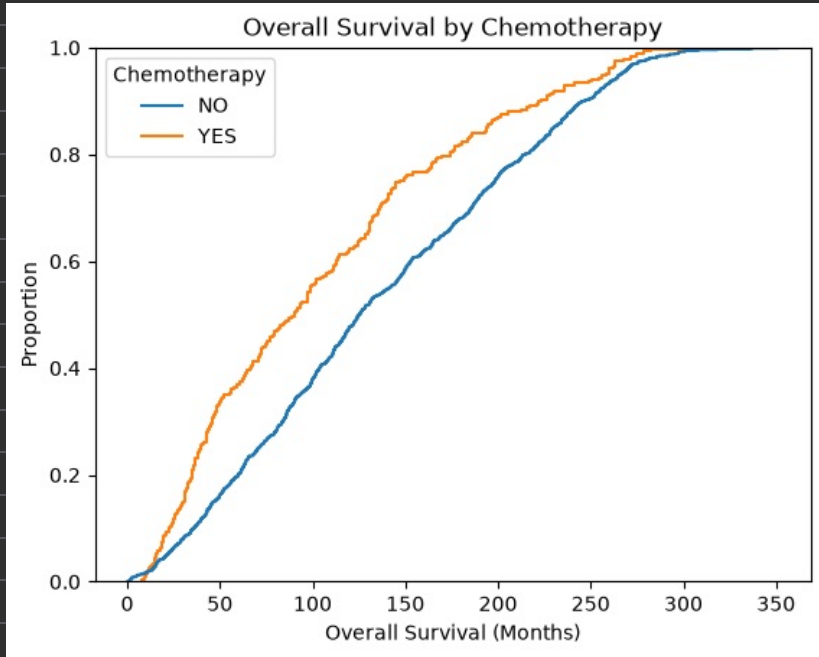
$$x = -\ln \left(\frac{1-u}{u} \right)$$

$$\therefore F^{-1}(u) = \ln \left(\frac{u}{1-u} \right), \quad 0 < u < 1$$

6 a)



b)



patients who receive chemotherapy tend to have shorter overall survival times. The ECDF for the chemotherapy group is generally shifted to the no chemotherapy group. Meaning that for a given cumulative probability, chemotherapy

patients reach that probability at a lower survival time. $\therefore$ conditional on a patient receiving chemotherapy we would predict a shorter survival time.

c) we cannot determine from this plot alone whether chemotherapy is an effective treatment. Although patients who received chemotherapy have shorter observed survival times this is an association and does not establish that chemotherapy causes shorter survival. Other factors such as disease severity, age or cancer characteristics, may effect both whether a patient receives chemotherapy and their survival.